WheatScan: AI-Based Wheat Disease Detection System

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Problem Statement

Yield & Quality Loss

Wheat diseases significantly reduce crop yield and quality, threatening food security and farmer livelihoods.

Expert Knowledge Gap

Manual identification requires expert knowledge that most farmers do not have access to.

Limited Specialist Access

Farmers often lack immediate access to agricultural specialists, especially in remote areas.

Delayed Diagnosis

Delayed diagnosis leads to increased crop damage and greater economic losses over time.

Proposed Solution

WheatScan is a fast, accessible, and practical disease detection system powered by deep learning — designed to put expert-level diagnosis in the hands of every farmer.

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Upload Image

Farmer captures and uploads a photo of the affected wheat leaf.

AI Detection

Deep learning model identifies the disease with high accuracy.

Severity Prediction

Predicts the severity level of the detected disease.

Treatment Guidance

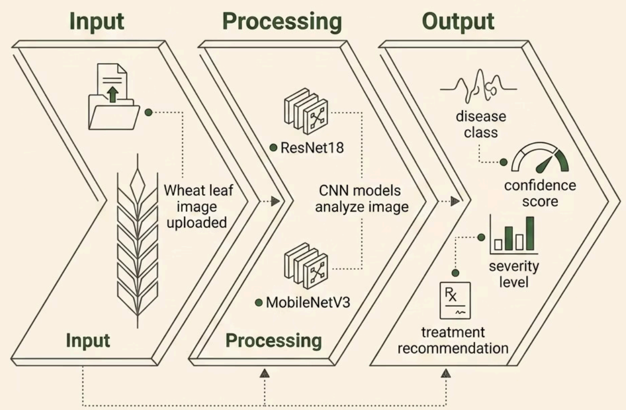
Provides actionable treatment recommendations instantly.

Technical Approach

Dataset & Architecture

The system is trained on the **PlantVillage Dataset**, covering **15 disease classes plus healthy** samples. A separate severity classification model handles disease intensity assessment.

Image-based classification using Convolutional Neural Networks (CNNs) enables the model to learn visual disease patterns directly from leaf images.



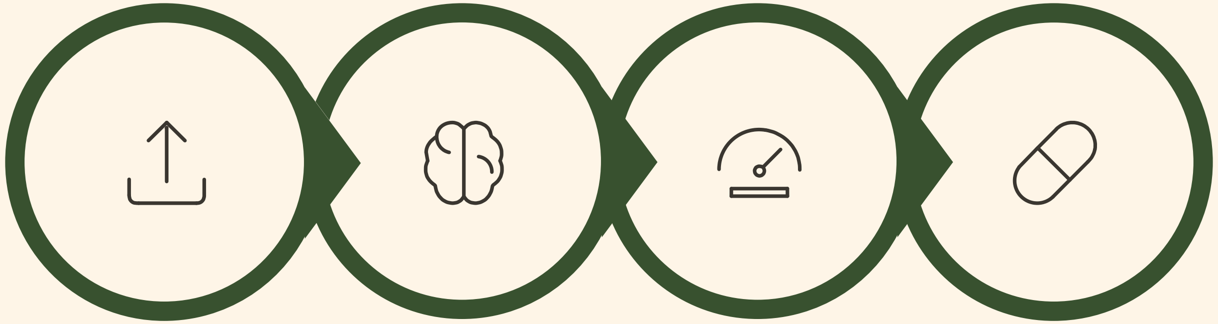
ResNet18

Deep residual network for robust feature extraction

MobileNetV3

Lightweight architecture optimized for efficiency

System Pipeline



Upload Image

Model Predicts disease

Model predicts severity

Treatment Advice

The end-to-end pipeline delivers results in seconds, giving farmers immediate, actionable insights from a single photo.

🌾 Disease

Brown Rust

📊 Confidence

~90%

⚠️ Severity

Moderate

💊 Treatment

Recommendations provided

Challenges & Results

Challenges Faced

Class Imbalance

Uneven distribution of samples across disease classes in the dataset.

Visually Similar Diseases

Some diseases present nearly identical visual symptoms on leaves.

Limited Data

Scarce samples for certain disease classes affected model generalization.

Key Results

91%

Disease Classification

Accuracy on disease class prediction across 15+ classes

87%

Severity Prediction

Accuracy in assessing disease severity levels

High accuracy achieved on major diseases with reliable predictions in most cases. Slight confusion observed in visually similar disease classes.

Future Work & Conclusion

Future Improvements

Conclusion

WheatScan delivers an **end-to-end AI system** for wheat disease detection — combining disease identification, severity assessment, and treatment guidance in a **user-friendly interface** designed for practical, real-world use.

01

Larger & Balanced Dataset

Expand training data to improve coverage and reduce class imbalance.

02

More Disease Classes

Add additional wheat diseases to broaden system applicability.

03

Mobile Application

Deploy as a mobile app for on-field use by farmers.

04

Real-World Field Testing

Validate system performance under actual farming conditions.

References

Mohanty, Hughes & Salathé (2016)

Using Deep Learning for Image-Based Plant Disease Detection

Ferentinos (2018)

Deep Learning Models for Plant Disease Detection

Too, E. C. et al. (2019)

Fine-Tuning Deep Learning Models for Plant Disease Identification

Hughes & Salathé (2015)

PlantVillage Dataset

All research and datasets referenced were instrumental in designing and validating the WheatScan system architecture and methodology.